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Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2025

Time: 03 Hours

Code: 100209 ***Subject: Basic Electronics*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[14]**

(a) When a pure semiconductor is heated, its resistance will
(i) Goes up (iii) Goes down
(ii) Remains the same (iv) None

(b) The random motion of holes and free electrons due to thermal agitation is called
(i) Diffusion (iii) Pressure
(ii) Ionisation (iv) None of these

(c) A zener diode is used as
(i) An amplifier (iii) A voltage regulator
(ii) A rectifier (iv) A multivibrator

(d) The ripple factor of a half-wave rectifier is
(i) 2 (iii) 1.21
(ii) 0.48 (iv) None

(e) The number of depletion layers in a transistor is
(i) Four (iii) Three
(ii) Two (iv) One

(f) In a transistor, I_C=100~mA and I_E=100.2~mA. The value of β is
(i) 100 (iii) 1
(ii) 50 (iv) 200

(g) A non inverting closed loop op-amp circuit generally has a gain factor
(i) Less than one (iii) Zero
(ii) Greater than one (iv) Equal to one

(h) If ground is applied to the (+) terminal of an inverting op-amp, the (-) terminal will
(i) Not need an input resistor (iii) Have high reverse current
(ii) Be virtual ground (iv) Not invert the signal

(i) The binary code of (21.125) is
(i) 10101.001 (iii) 10100.001
(ii) 10101.010 (iv) 10100.111

(j) Which of the following gate is a two-level logic gate
(i) OR gate (iii) NAND gate
(ii) EXCLUSIVE OR gate (iv) NOT gate

Q.2 Solve both questions : **[14]**

(a) Explain the working of p-n junction diode and draw its V-I Characteristics.

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(b) Draw a neat circuit diagram of bridge rectifier and explain its operation with output waveforms. Derive the average value of current and voltage.

Q.3 Solve both questions : **[14]**

(a) Explain working principle of Depletion type MOSFET (n-channel). Draw & Explain its characteristics.

(b) Explain the working principle and characteristics of a full-wave rectifier. Derive the expressions for output voltage and ripple factor for a full-wave rectifier.

Q.4 Solve both questions : **[14]**

(a) Draw the block diagram of Op-Amp and list all the ideal characteristics of op-amp.

(b) Explain the working of op-amp as an Integrator and derive its output equation.

Q.5 Solve both questions : **[14]**

(a) Draw the circuit of NPN transistor in common base configuration and discuss its working. Draw input-output characteristic.

(b) Explain the output and transfer characteristics of a JFET. How does the gate-source voltage (V_{GS}) affect the drain current (I_D)?

Q.6 Solve both questions : **[14]**

(a) Convert the following:
(i) $(53.625)_{10}$ to $(?)_2$
(ii) Find the base x if $(211)_x=(152)_8$
(iii) Subtract using 1's complement: (10111) − (110011)
(iv) Find the 1's and 2's compliment of $(010100)_2$

(b) Discuss the working of basic logic gates: NAND, NOR, XOR. Derive the truth tables and Boolean expressions for each gate.

Q.7 Solve both questions : **[14]**

(a) Explain the concept of load line analysis in a BJT. Derive the expressions for the load line and operating point. Explain their significance in amplifier design.

(b) Explain the difference between intrinsic and extrinsic semiconductors. How does the addition of impurities affect the electrical properties of semiconductors?

Q.8 Solve both questions : **[14]**

(a) Discuss the working principle of clipping circuits. What are the different types of clipping circuits?

(b) Explain the concepts of static and dynamic resistance in a diode. Derive expressions for both and discuss their significance in diode operation.

Q.9 Write short notes on any two of the following: **[14]**

(a) Solar cell

(b) Basic logic gates, their symbols and truth tables

(c) Clipping and Clamping circuits

(d) Transistor biasing techniques

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